

SYSTEM OVERVIEW & DESIGN SPECIFICATION:

This drawing package details the electrical schematics, power distribution, co-processor pulse interfaces, remote field I/O, and motion safety interlocks for the 3-axis automated electronics dispensing machine (Narit Smart Vending Machine).

Operating Temp: 0 - 50 °C
Rated Voltage: 220VAC 50Hz
Rated Power: 650W Max

1. POWER DISTRIBUTION

AC Mains, MCB, Filter, PSUs

- AC Mains 220VAC 50Hz Inlet
- 2-Pole MCB C16 Circuit Breaker
- CW4L2-20A-S Noise Filter & SPD
- PSU 1: 60VDC 6.7A (Motors X/Y)**
- PSU 2: 24VDC 5A (Control & Z)**
- PSU 3: 5VDC 3A (CM4 Isolated)**
- KM1 Safety Contactor & Cutoff**
- E-Stop Hardwired Safety Loop

Refer to SHEET 2 for detailed wiring.

2. MAIN CONTROLLER

Cytron IRiV PiControl CM4

- Raspberry Pi CM4 (4GB RAM, 32GB eMMC)
- eth0: MGMT LAN (192.168.70.80 / UI)
- eth1: OT LAN (10.0.0.2 Modbus Master)
- USB4: ST-LINK V3 Serial (115200 8-N-1)**
- Local DI0-DI1: X/Y Drive Alarm (ALM)**
- Local DI2-DI3: X/Y In-Position (PEND)**
- Local DO0: KM1 Coil Control & Reset**
- Software Watchdog: Auto-disarm 350ms**

Refer to SHEET 3 for detailed wiring.

3. REMOTE FIELD I/O

Cytron IRiV IO Modbus TCP

- Modbus TCP Slave (10.0.0.10:502)
- 11 x Digital Inputs (24V Opto):
 - DI0-DI5: Limits X/Y/Z Min & Max
 - DI6: Z Home Reference Sensor
 - DI7: Drop Alignment Sensor
 - DI8: Drop Beam (Omron E3Z)
 - DI9: Pickup Door (Omron E3Z)
 - DI10: KM1 E-Stop Loop (NC)
- 4 x Digital Outputs (Pilot Lights/Relay)

Refer to SHEET 4 for detailed wiring.

4. MOTION CO-PROCESSOR

STM32 NUCLEO-G491RE + NMOS

- ARM Cortex-M4 @ 170MHz
- Hardware Timer Compare Pulse Gen
- X: PA8 (TIM1_CH1) / PB0 (GPIO)**
- Y: PA9 (TIM1_CH2) / PB1 (GPIO)**
- Z: PA5 (TIM2_CH1) / PB2 (GPIO)**
- 6-Channel NMOS Open-Drain Board
- Common Anode V-PULSE (+24VDC)**
- Safe-Link Protocol v3 / Watchdog**

Refer to SHEET 5 for detailed wiring.

5. DRIVES & ACTUATORS

HBS860H, DM542, Steppers

- Drive X: HBS860H Closed-Loop
 - Motor: 86HBS85 (8.5 N.m)
- Drive Y: HBS860H Closed-Loop
 - Motor: 86HBS85 (SFU1605)
- Drive Z: DM542 Microstepping
 - Motor: NEMA 17 Pusher
- Differential Optical Encoders
- V-PULSE 24V Opto Isolation**

Refer to SHEET 6 for detailed wiring.

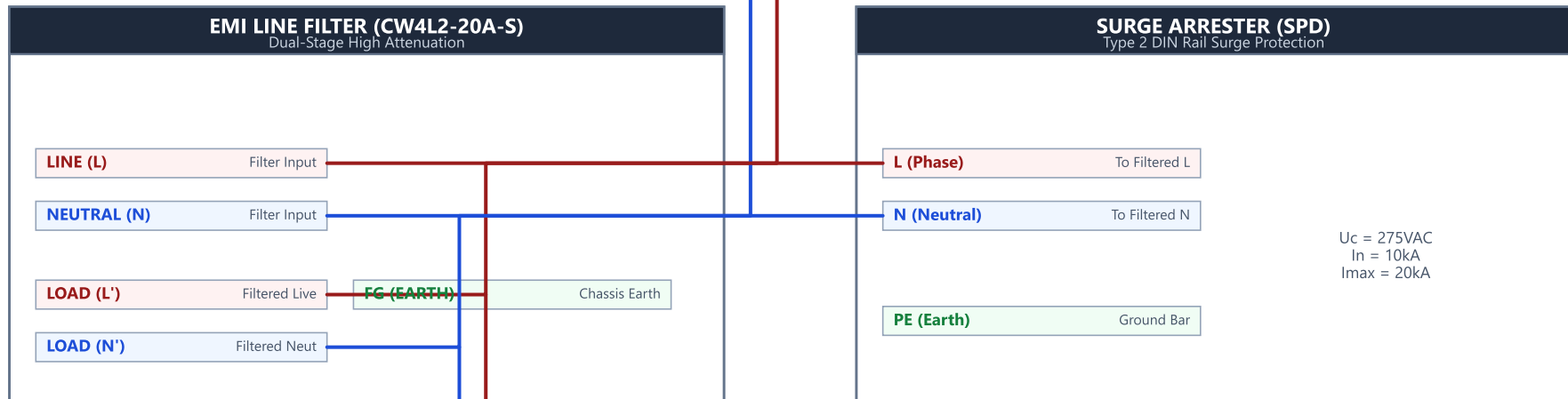
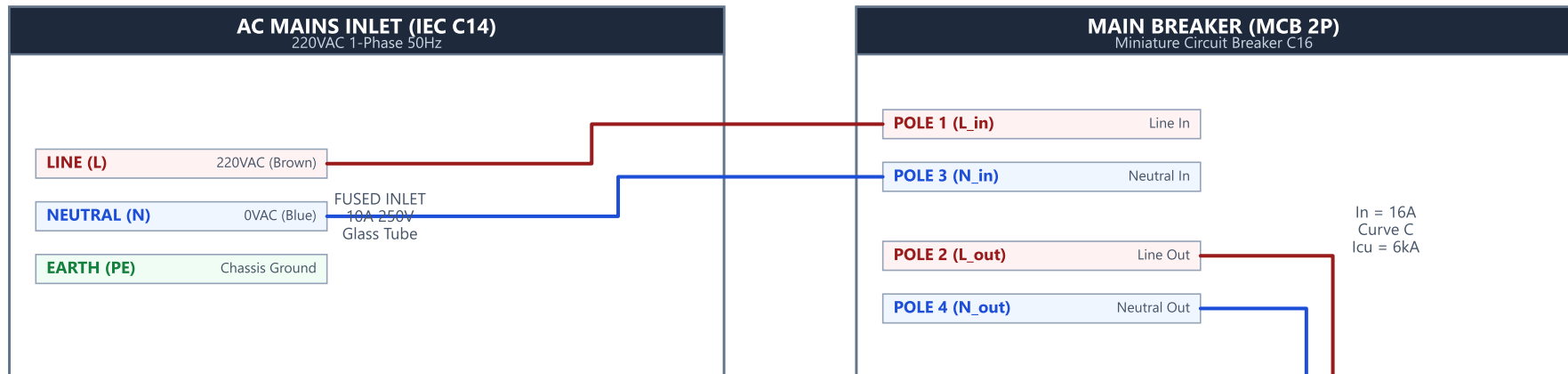
MASTER INTERCONNECTION & CABLE SCHEDULE

Cable IDs, Signal Types, Terminations, and Specifications

CABLE ID	FROM (SOURCE)	TO (DESTINATION)	SIGNAL / FUNCTION	CONDUCTORS / GAUGE	ROUTING / REMARKS
W-01	AC Inlet C14 / Main Cord	MCB 2P 16A Input (1, 3)	220VAC Mains Power	3 x 2.5 mm² (L, N, PE)	Mains Cable Raceway
W-02	MCB Output (2, 4)	CW4L2-20A-S Noise Filter In	Protected AC Mains	2 x 2.5 mm² (L, N)	Cabinet Power Duct
W-03	Noise Filter Load (L', N')	AC Terminal Bus TB-L / TB-N	Clean Filtered AC Bus	2 x 2.5 mm² (L, N)	DIN Rail Distribution
W-04	TB-L / TB-N / TB-PE	PSU1, PSU2, PSU3 AC Inputs	Parallel DC PSU Feeds	3 x (3 x 1.5 mm²)	DIN Rail Jumper Bars
W-05	PSU 1 (+60V, 0V)	KM1 Contactor (1-2, 3-4)	High Voltage Motor DC	2 x 2.5 mm² (Red, Blk)	Direct to KM1 Contacts
W-06	KM1 Contactor Out	Drivers X & Y AC/AC Terminals	Motor Bus (+60VDC)	2 x 2.5 mm² (Red, Blk)	Separated from Signal
W-07	PSU 2 (+24V, 0V)	PiControl, IRiV IO, Sensors	Control & Logic Bus	2 x 1.5 mm² (Orn, Blk)	Main DC Control Duct
W-08	PSU 3 USB-C (5V 3A)	IRiV PiControl CM4 USB-C	Isolated CM4 Power	Official 18AWG Type-C	Dedicated Short Run
W-09	PiControl eth1 (OT LAN)	IRiV IO Modbus RJ45 Port	Modbus TCP 10.0.0.10:502	Cat6 UTP Patch Cable	Industrial Ethernet
W-10	PiControl USB4 Port	NUCLEO-G491RE ST-LINK VCP	Safe-Link Protocol v3	Shielded USB-A to Micro-B	Ferrite Choke Protected
W-11	PiControl DO0 (GPIO 23)	KM1 Contactor Coil A1/A2	Safety Cutoff / 3s Reset	2 x 0.75 mm² Twisted	Series with E-Stop NC
W-12	KM1 Contactor Aux NC	IRiV IO DI10 Terminal	Hardware State Feedback	2 x 0.5 mm² Shielded	Fail-Safe NC Loop
W-13	STM32 Morpho Headers	6-Ch NMOS Board Gates	3.3V Timer Pulse/Dir	Ribbon Cable 10-pin	< 150mm Short Run
W-14	NMOS Drains & V-PULSE	Drivers X, Y, Z (PUL, DIR)	24V Opto-Isolated Drive	6 x Twisted Pairs 0.5mm²	Separated from Motor 60V
W-15	Drivers X & Y Output	86HBS85 Motors A/B & Encoders	Motor Coils & Differential	Shielded 4-core + Enc	Grounded at Cabinet PE
W-16	Field Proximity & Optical	IRiV IO DI0 - DI9 Terminals	24V Limit & Sensor Inputs	Shielded 3-core 0.5mm²	Bottom Raceway Duct

AC MAINS 220VAC INLET & NOISE SUPPRESSION

Single-Phase 220V 50Hz, Circuit Protection, and EMI Filtering



AC DIN RAIL DISTRIBUTION TERMINAL BLOCKS

Screw Clamp Terminal Blocks with Jumper Bars (1-in 4-out)



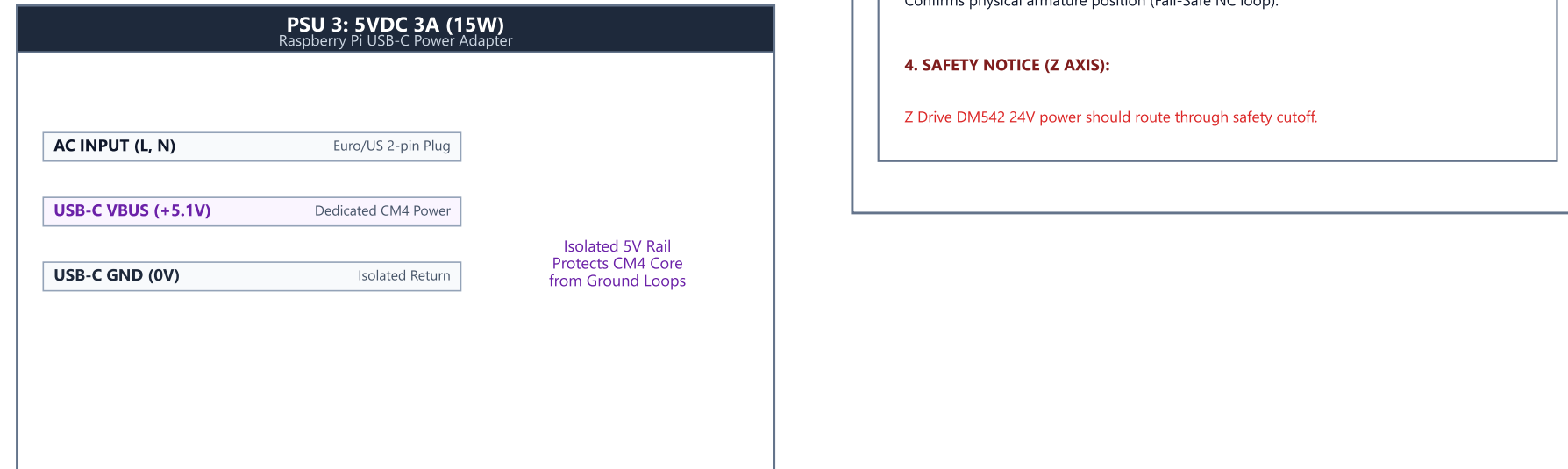
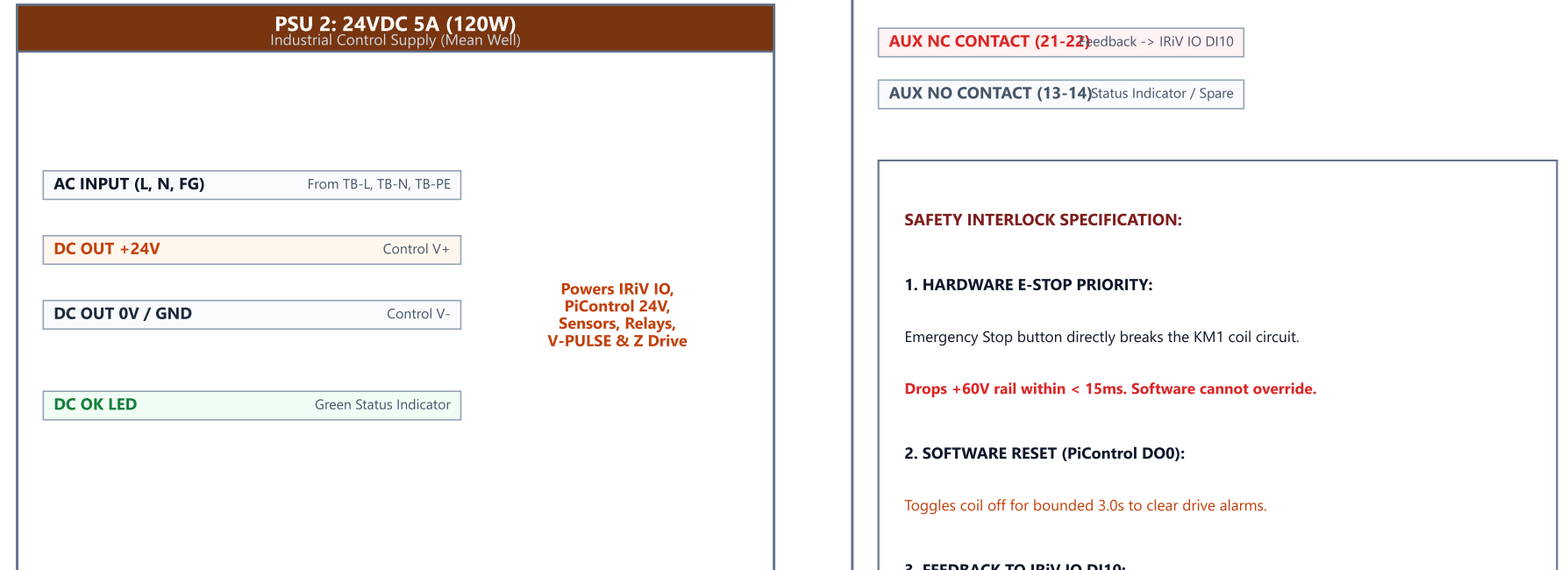
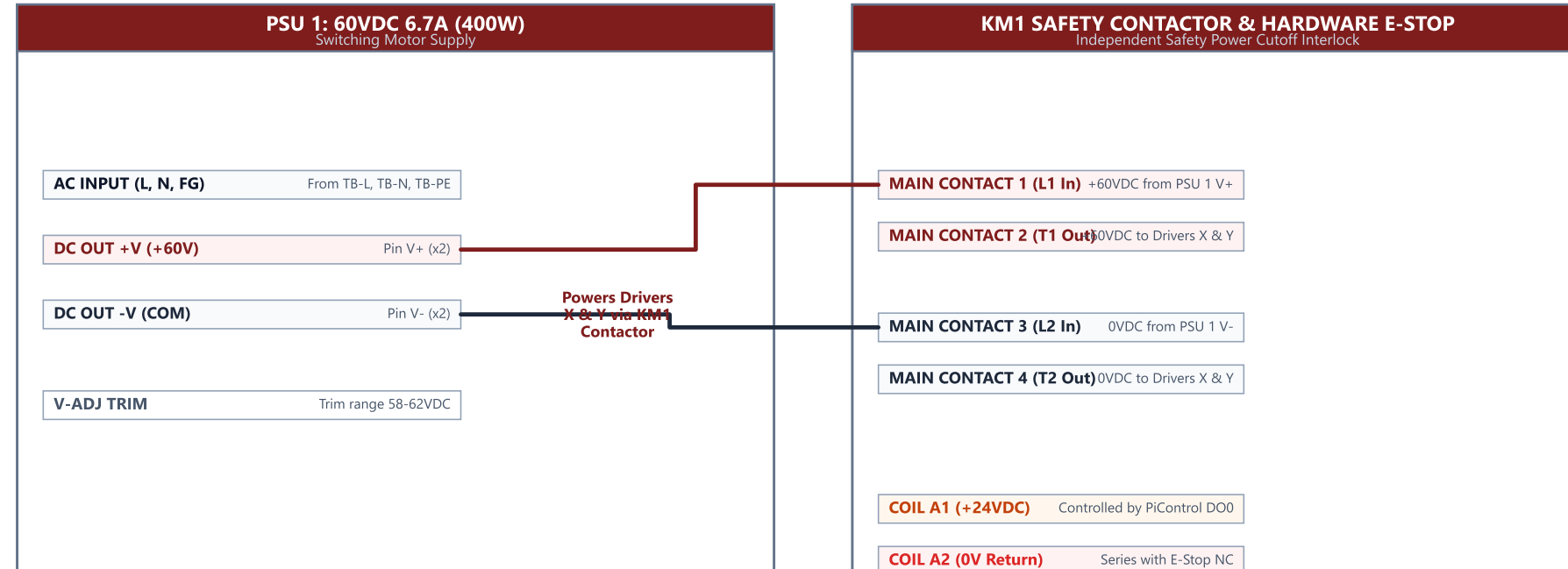
AC BUS FEED ALLOCATIONS:

- Branch 1: PSU 1 (60V 6.7A Motor Supply)
- Branch 2: PSU 2 (24V 5A Control Supply)
- Branch 3: PSU 3 (5V 3A USB-C CM4 Supply)
- Branch 4: Spare DIN Rail Terminal

Wiring Standard: IEC 60204-1
AC Phase: Brown 2.5 mm²
Neutral: Light Blue 2.5 mm²
Earth: Green/Yellow 2.5 mm²

DC POWER SUPPLIES & KM1 SAFETY POWER INTERLOCK

Multi-Voltage DC Power Rails with Hardware Emergency Contactor Cutoff



PROJECT: NARIT SMART VENDING MACHINE | SECTION 2: MAIN CONTROLLER (CYTRON IRIV PICONTROL CM4)

SECTION: MAIN CONTROLLER | TOR: 00-TOR-Vending-J69-290

Cytron IRiV PiControl CM4 WIRELESS CONTROLLER

Raspberry Pi CM4 (4GB RAM, 32Gb eMMC, Wi-Fi, Dual Ethernet, Isolated GPIO)

1. SYSTEM POWER & DUAL GIGABIT ETHERNET INTERFACES:

DC 24V IN (+)	Auxiliary 24VDC Terminal	USB-C VBUS (5V)	CM4 Core Primary Power	eth0: MGMT LAN	192.168.70.80 / Flask REST
DC 0V IN (-)	Field Ground 0VDC	PWR / ACT LED	System Status Indicators	eth1: OT LAN	10.0.0.2 / Modbus Master

2. USB 2.0 PERIPHERAL PORTS (DEVICE INTERFACES):

USB 1: QR SCANNER	2D Barcode / Payment QR CDC	USB 2: WEB CAMERA	FHD 1080p UVC Video Stream
USB 3 / 3.5mm AUDIO	Chime / Audio Voice Guidance	USB 4: ST-LINK VCP	To NUCLEO-G491RE (115200 8-N-1)

3. LOCAL OPTO-ISOLATED GPIO TERMINAL BLOCK (FAIL-SAFE CHANNELS):

D10 (GPIO 13)	X_DRIVE_ALM	Driver X Alarm (Fail-Safe)	Active HIGH / Trips on open-circuit
D11 (GPIO 17)	Y_DRIVE_ALM	Driver Y Alarm (Fail-Safe)	Active HIGH / Trips on open-circuit
D12 (GPIO 27)	X_PEND	Driver X In-Position	Position-End verification signal
D13 (GPIO 22)	Y_PEND	Driver Y In-Position	Position-End verification signal
DO0 (GPIO 23)	XY_DRIVE_POWER_KM1	KM1 Safety Contactor Driver	Hardware 3.0s reset pulse & safety coil
DO1 (GPIO 24)	RELAY_AUX_1	Auxiliary Output 1	General purpose 24V SSR driver
DO2 (GPIO 25)	RELAY_AUX_2	Auxiliary Output 2	General purpose 24V SSR driver
DO3 (GPIO 16)	RELAY_AUX_3	Auxiliary Output 3	General purpose 24V SSR driver

SYSTEM DAEMONS & IPC CONTROL GATES:

- Daemon 1: narit-vending-web-iriv.service (Flask HMI Port 80, Touchscreen Kiosk UI)
- Daemon 2: narit-vending-controller-iriv.service (Hardware Safety Interlock Engine)
- IPC Bus: Unix Domain Socket /run/narit-vending/ctrl.sock (Non-blocking JSON)
- Modbus TCP Master: Polls IRiV IO (10.0.0.10:502) every 20ms; Disarms if stale > 350ms
- Safe-Link Protocol v3: USB Serial to STM32 (115200 8-N-1); Heartbeat timeout 500ms
- Cloud Broker: MQTT Hub (broker.emqx.io:1883) for telemetry, status and inventory
- Persistent Storage: SQLite database + machine_config.iriv.json configuration

PERIPHERAL MODULES & FIELD CABLE TERMINATIONS

Scanner, Camera, Audio, Ethernet, and ST-LINK Connections

2D QR CODE SCANNER

Barcode & Payment Scanner

VBUS (+5VDC)	USB Power Pin 1
D- / D+ (DATA)	USB 2.0 Differential
GND (0V RETURN)	USB Ground Pin 4
BEEPER / LED	Scan Confirmation

FHD WEB CAMERA

Dispense / Pickup Vision

VBUS (+5VDC)	USB Power Pin 1
D- / D+ (UVC)	1080p Video Stream
GND (0V RETURN)	USB Ground Pin 4
INDICATOR LED	Active Stream Light

AUDIO AMPLIFIER & SPEAKER

Voice Guidance / Chime

AUDIO L / R IN	3.5mm Stereo Jack
POWER (+5V / 0V)	USB Power Supply
SPEAKER OUT (3W)	4-Ohm Transducer
VOLUME POT	Hardware Gain Adjust

ST-LINK V3 SERIAL VCP

To NUCLEO-G491RE

USB CDC ACM	/dev/serial/by-id/...
BAUDRATE	115200 8-N-1
PROTOCOL	Safe-Link v3 Binary
HEARTBEAT	Timeout = 500 ms

ETHERNET DUAL-SUBNET ARCHITECTURE

Physical Network Separation (Air-Gapped Management vs OT)

SUBNET 1: MANAGEMENT LAN (eth0)

- IP: 192.168.70.80 / 24 | Gateway: 192.168.70.1
- Function: Kiosk Touchscreen HMI, Web API, Cloud MQTT, Remote SSH

SUBNET 2: OPERATIONAL TECHNOLOGY LAN (eth1)

- IP: 10.0.0.2 / 24 | Target: 10.0.0.10 (Cytron IRiV IO Modbus)
- Function: High-speed cyclic I/O polling (20ms interval)

- Safety: Isolated industrial broadcast domain; prevents network collisions.

Cytron IRiV IO CONTROLLER (MODBUS TCP)

Industrial Remote Field I/O (11 DI, 4 DO, 24VDC Opto-Isolated)

MODBUS TCP RJ45

IP: 10.0.0.10:502 / ID: 255

POWER (+24V, 0V, S/S)

S/S tied to 0V (PNP Mode)

DIGITAL INPUTS (DI0 - DI10) — 24VDC OPTO-ISOLATED:

DI0: x_head_limit

X Min Limit / Home Sensor

Normally Closed (NC)

Inductive Proximity

DI1: x_tail_limit

X Max Limit Sensor

Normally Closed (NC)

Inductive Proximity

DI2: y_head_limit

Y Min Limit / Drop Level

Normally Closed (NC)

Inductive Proximity

DI3: y_tail_limit

Y Max Limit / Top Level

Normally Closed (NC)

Inductive Proximity

DI4: z_head_limit

Z Min Limit / Retract End

Normally Closed (NC)

Inductive Proximity

DI5: z_tail_limit

Z Max Limit / Extend End

Normally Closed (NC)

Inductive Proximity

DI6: z_home

Z Home Reference Sensor

Normally Open (NO)

Inductive Proximity

DI7: product_drop_parking

Drop Alignment Proximity

Normally Open (NO)

Inductive Proximity

DI8: product_drop_sensor

Completed Drop Sensor

Beam Interrupted

Omron E3Z-D81 Optical

DI9: product_pickup_sensor

Door Access Pickup Sensor

Door Flap Opened

Omron E3Z-D81 Optical

DI10: estop / KM1_FEEDBACK

Safety Contactor Aux NC

Fail-Safe NC Loop

KM1 Contactor Aux

DIGITAL OUTPUTS (DO0 - DO3) — 24VDC SSR / RELAYS:

DO0: ready

Green Pilot Light

Machine Ready Indication

24V / 0.5A SSR

DO1: moving

Yellow Pilot Light

Carriage Moving Indication

24V / 0.5A SSR

DO2: alarm

Red Light & Audible Buzzer

Fault / Emergency Indication

24V / 0.5A SSR

DO3: dispense

Solenoid Drop Gate Relay

Timed Pulse 500ms Trigger

24V / 0.5A SSR

FIELD SENSOR WIRING SCHEMATICS & SAFETY LOGIC

Inductive Proximity, Optical Sensors, and Safety Relay Loops

3-WIRE INDUCTIVE PROXIMITY SENSORS (PNP NO/NC)

Connection Diagram for Limit Sensors X, Y, Z (DI0 - DI7)

INDUCTIVE
SENSOR
(PNP)Shield Grounded
at Cabinet PE

BROWN (BN)

+24VDC Power (From PSU 2 Rail)

BLUE (BU)

0VDC / COM Return (From PSU 2 Rail)

BLACK (BK)

Signal Output -> IRiV IO (DI0 - DI7)

S/S TERMINAL

Must be jumpered to 0V for PNP inputs

OMRON E3Z-D81 PHOTOELECTRIC SENSORS

Connection Diagram for Drop (DI8) & Pickup (DI9) Verification

BROWN (BN)

+24VDC Supply Rail

BLUE (BU)

0VDC Power Return

BLACK (BK)

Output -> DI8 (Drop Beam)

WHITE (WH)

Output -> DI9 (Pickup Flap)

SENSITIVITY

Single-turn trimmer adjust for beam distance

CRITICAL SAFETY HOLD POINTS & INVARIANTS

Hardware Interlocks that MUST be verified before operation

1. FAIL-SAFE NC LIMIT CONFIGURATION:

Limit sensors DI0-DI5 use Normally Closed contacts. A broken cable trips E-Stop instantly.

2. DI10 POLARITY VERIFICATION (E-STOP LOOP):

DI10 active LOW (NC loop). Software blocks motion if contactor drops or loop opens.

3. MODBUS TIMEOUT GUARD: Controller disarms all axes if poll response stale > 350ms.

STM32 NUCLEO-G491RE MOTION CO-PROCESSOR

ARM Cortex-M4 @ 170MHz (Hardware Timer Output Compare Pulse Engine)

ST-LINK V3 USB (VCP)

LPUART1 PA2/PA3 @ 115200 8-N-1

PROTOCOL: SAFE-LINK v3

Binary Frame with CRC & Watchdog

HARDWARE TIMER & GPIO PIN MAPPING (NUCLEO-64 MORPHO & ARDUINO HEADERS):

X_STEP

PA8 (TIM1_CH1 (AF6))

Morpho CN10 pin 23

Arduino D7

X_DIR

PB0 (GPIO Output)

Morpho CN7 pin 34

Arduino A3

Y_STEP

PA9 (TIM1_CH2 (AF6))

Morpho CN10 pin 21

Arduino D8

Y_DIR

PB1 (GPIO Output)

Morpho CN10 pin 24

(No Arduino)

Z_STEP

PA5 (TIM2_CH1 (AF1))

Morpho CN10 pin 11

Arduino D13

Z_DIR

PB2 (GPIO Output)

Morpho CN10 pin 22

(No Arduino)

MCU_GND

GND (Reference 0V)

Morpho CN10 pin 20

CN7 pin 20

CRITICAL PIN MISWIRING WARNINGS (DO NOT USE OLD NUCLEO-144 PINS):

- X_DIR (PB0): DO NOT plug into CN10-31! On Nucleo-64 that pin is PB3. Connect to CN7 pin 34.
- Y_DIR (PB1): DO NOT plug into CN10-7! On Nucleo-64 that is AVDD 3.3V power (locks DIR HIGH). Connect to CN10 pin 24.
- Z_DIR (PB2): DO NOT plug into CN10-15! On Nucleo-64 that is PA7. Connect to CN10 pin 22.
- GROUND: DO NOT connect GND to CN10-5, 17, 27! Those are GPIO pins. Use ONLY CN10 pin 20 or CN10 pin 9.
- Z_STEP (PA5) CONFLICT: PA5 shares user LED LD2. Firmware MUST have all LED blink code disabled.

Failure to follow this mapping will cause false motion steps, frozen direction, or damaged MCU pins.

FIRMWARE OPERATIONAL SAFETY LIMITS:

- Boot Invariant: Starts DISARMED, timers stopped, STEP & DIR lines held LOW.
- Frequency Range: Hardware Timer output compare clamped strictly between 10 Hz and 50,000 Hz.
- Block Batch Limit: Maximum 1,000,000 steps per block command.
- Axis Mutual Exclusion: Only 1 motion axis actively steps at any given moment.

6-CHANNEL LOGIC-LEVEL NMOS SINK BOARD

Open-Drain Level Translation (3.3V MCU Logic -> 24V Optocoupler Sink)

OPTOCOUPLER DRIVE CIRCUIT TOPOLOGY (PER CHANNEL):

+24VDC (V-PULSE Bus) -----+--> Driver PUL+ / DIR+ (Anode)

| (Internal Optocoupler LED)

STM32 GPIO ----[100 Ohm]----+--- Gate (Qn) +--> Driver PUL- / DIR- (Cathode)

| ^

[10k Ohm] Drain (Qn)

MCU GND -----+----- Source (Qn) ----> Field 0VDC / 0V-SIGNAL

6-CHANNEL NMOS PIN ASSIGNMENTS:

CH 1: X_PULSE

Gate: STM32 PA8 (CN10-23)

Drain -> Driver X PUL-

Low-Side Sink

CH 2: X_DIR

Gate: STM32 PB0 (CN7-34)

Drain -> Driver X DIR-

Low-Side Sink

CH 3: Y_PULSE

Gate: STM32 PA9 (CN10-21)

Drain -> Driver Y PUL-

Low-Side Sink

CH 4: Y_DIR

Gate: STM32 PB1 (CN10-24)

Drain -> Driver Y DIR-

Low-Side Sink

CH 5: Z_PULSE

Gate: STM32 PA5 (CN10-11)

Drain -> Driver Z PUL-

Low-Side Sink

CH 6: Z_DIR

Gate: STM32 PB2 (CN10-22)

Drain -> Driver Z DIR-

Low-Side Sink

V-PULSE BUS (+24VDC)

Common Anode Rail to all PUL+ & DIR+

0V-SIGNAL / COMMON GND

Tied to all NMOS Sources, MCU GND, and PSU 2 0V

COMPONENT SPECIFICATIONS & ENGINEERING NOTES:

- MOSFET: Logic-Level N-Channel (2N7002 / AO3400 / IRF520 / BSS138), Vds >= 50V, Vgs(th) <= 2.0V.
- Gate Resistor: 100-Ohm series damping resistor prevents ringing on high-frequency edges.
- Pulldown Resistor: 10k-Ohm pulldown guarantees off-state when MCU pins are in reset/tri-state.

PROJECT: NARIT SMART VENDING MACHINE | SECTION 5: STEPPER MOTOR DRIVERS & ACTUATORS WIRING (AXES X, Y, Z)

SECTION: DRIVES & ACTUATORS | TOR: 00-TOR-Vending-J69-290

AXIS X: HORIZONTAL CARRIAGE DRIVE

Leadshine HBS860H Closed-Loop Driver + 86HBS85 NEMA 34 Motor

DRIVER X: HBS860H (CLOSED-LOOP)

Peak 8.2A, Continuous 5.6A, Microstep 1600

AC / AC (V+ / V-) +60VDC & 0V from KM1 Contactor

PUL+ (V-PULSE) +24VDC Common Anode Bus PUL- (STEP) From NMOS Ch1 (PA8)

DIR+ (V-PULSE) +24VDC Common Anode Bus DIR- (DIR) From NMOS Ch2 (PB0)

ALM+ / ALM- Alarm Opto -> PiControl DI0 PEND+ / PEND- In-Pos Opto -> PiControl DI2

COIL A+ / A- Phase A (Red / Blue) COIL B+ / B- Phase B (Black / Green)

EA+ / EA- Channel A Differential EB+ / EB- Channel B Differential

VCC / EGND Enc Power (+5V Red / 0V Wht)

DIP SW1-SW4 Current: SW1-SW4 = ON, ON, ON, ON (Peak 8.2A)

DIP SW5-SW8 Microstep: SW5-SW8 = ON, OFF, ON, ON (1600 steps/rev)

RS232 Tuning Port: ProTuner parameter configuration interface

MOTOR X: 86HBS85 (NEMA 34)

8.5 N.m Closed-Loop Stepper + 1000 CPR Encoder

COIL LEADS (4-PIN) Red: A+ | Blu: A- | Blk: B+ | Grn: B-

ENCODER (6-PIN) EA+, EA-, EB+, EB-, VCC(+5V), EGND(0V)

SHAFT & TRANSMISSION 14mm Keyed Shaft -> HTD 5M 25mm Timing Belt

KINEMATIC RATIO 78.43 steps/mm @ 1600 microstep (Pitch Dia 63.66mm)

TRAVERSE PERFORMANCE Rapid: 450 mm/s | Accel: 800 mm/s² | Dual HGR20 Guides

Closed-loop feedback eliminates stalling. Trips ALM on > 1000 count lag.

AXIS Y: VERTICAL ELEVATOR DRIVE

Leadshine HBS860H Closed-Loop Driver + 86HBS85 NEMA 34 Motor

DRIVER Y: HBS860H (CLOSED-LOOP)

Peak 8.2A, Continuous 5.6A, Microstep 1600

AC / AC (V+ / V-) +60VDC & 0V from KM1 Contactor

PUL+ (V-PULSE) +24VDC Common Anode Bus PUL- (STEP) From NMOS Ch3 (PA9)

DIR+ (V-PULSE) +24VDC Common Anode Bus DIR- (DIR) From NMOS Ch4 (PB1)

ALM+ / ALM- Alarm Opto -> PiControl DI1 PEND+ / PEND- In-Pos Opto -> PiControl DI3

COIL A+ / A- Phase A (Red / Blue) COIL B+ / B- Phase B (Black / Green)

EA+ / EA- Channel A Differential EB+ / EB- Channel B Differential

VCC / EGND Enc Power (+5V Red / 0V Wht)

DIP SW1-SW4 Current: SW1-SW4 = ON, ON, ON, ON (Peak 8.2A)

DIP SW5-SW8 Microstep: SW5-SW8 = ON, OFF, ON, ON (1600 steps/rev)

Elevator Holding: Holding current maintains vertical level under load

MOTOR Y: 86HBS85 (NEMA 34)

8.5 N.m Closed-Loop Stepper + SFU1605 Ball Screw

COIL LEADS (4-PIN) Red: A+ | Blu: A- | Blk: B+ | Grn: B-

ENCODER (6-PIN) EA+, EA-, EB+, EB-, VCC(+5V), EGND(0V)

SHAFT & TRANSMISSION 14mm Shaft -> Direct Flexible Coupling to SFU1605

KINEMATIC RATIO 320.00 steps/mm @ 1600 microstep (Ball Screw Lead 5mm)

ELEVATOR PERFORMANCE Speed: 120 mm/s | Accel: 600 mm/s² | Direct Lift Carriage

Lifts Dispensing Basket to target product shelf slots (Levels 1 to 5).

AXIS Z: DISPENSER PUSHER DRIVE

Leadshine DM542 Driver + NEMA 17 V-Slot Mini Actuator

DRIVER Z: DM542 (MICROSTEPPING)

Peak 4.2A, 20-50VDC Supply

V+ / GND (24-48V) +24VDC & 0V from PSU 2 Control Rail

PUL+ (V-PULSE) +24VDC Common Anode Bus PUL- (STEP) NMOS Ch5 (PA5)

DIR+ (V-PULSE) +24VDC Common Anode Bus DIR- (DIR) NMOS Ch6 (PB2)

ENA+ / ENA- Enable Input (Not connected / enabled)

COIL A+ / A- Phase A Motor (Red / Blue) COIL B+ / B- Phase B (Black / Green)

DIP SW1-SW4 Current: SW1=ON, SW2=OFF, SW3=ON, SW4=OFF (Peak 2.0A)

DIP SW5-SW8 Microstep: SW5=ON, SW6=OFF, SW7=ON, SW8=ON (1600 step/rev)

Safety Recommendation: Route 24V supply through KM1

contactor to prevent unmonitored plunger motion during E-Stop.

MOTOR Z: NEMA 17 PUSHER

V-Slot Lead Screw Mini Actuator (150mm Stroke)

COIL LEADS (4-PIN) Red: A+ | Blu: A- | Blk: B+ | Grn: B-

ACTUATOR TYPE T8x8 Lead Screw (Pitch 2mm, 4 Starts = 8mm/rev)

KINEMATIC RATIO 200.00 steps/mm @ 1600 microstep (Lead 8.0mm/rev)

STROKE & TRAVEL Stroke: 150mm | Plunger pushes items onto basket tray

INTERLOCK RULE Must confirm Z_MIN (Retracted) before X/Y move

Hardware interlock prevents carriage collision while pusher is extended.